

Figure 1: Installation of BigBlueButton



Figure 2: Checking for errors in BigBlueButton

This process will take more than an hour to install ffmpeg, and it adds some modules into the kernel to support the audio, video, etc.

Installing BigBlueButton

Finally, install the BigBlueButton software using the following command:

```
$ sudo apt-get install bigbluebutton
```

The above command will install the entire library of software with all the necessary dependencies, and this will download as much as 400MB of software from the server. Figure 1 shows the installation of BigBlueButton.

Installation of demo software

If you want to see a demonstration of how BigBlueButton is used, you can install the software using the following command:

```
$ sudo apt-get install bbb-demo
```

If you want to remove this demo software, then use the following command:

```
$ sudo apt-get purge bbb-demo
```

Other settings

To enable the WebRTC (audio) for the server, the following

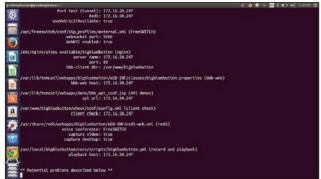


Figure 3: BigBlueButton in the local server

command can be used:

```
$ sudo bbb-conf --enablewebrtc
```

To make a clean restart of the BigBlueButton server, you can use the following commands:

```
$ sudo bbb-conf --clean
```

```
$ sudo bbb-conf --check
```

--clean will clear all the log files of BigBlueButton, and --check will search through and find any errors. See Figure 2 for the --check command. The IP address allocated to my server is 172.16.50.247 (it's a local IP). To check the status of my server, you can look at Figure 3.

Once the demo server is opened, just distribute the IP addresses so that the users can be connected to the IP and can join the session. In the first session, the browser checks for devices like microphones, speakers and webcams. The users have the option of joining the session only as listeners or both as listeners and speakers. In case users don't want to allow the browser to access a particular device, they can block that device within the browser.

Integrating BigBlueButton with Moodle (a case study)

Moodle is an open source learning management system (LMS) that comes with various plugins for additional functionalities. BigBlueButton also has two plugins—RecordingsBN and BigBlueButtonBN

BigBlueButtonBN: This plugin integrates Moodle and BigBlueButton, so that a teacher can create an activity within Moodle and the students can join the session. The teacher has total control of the session. The students can join the session only after the teacher joins the session. Also, the session can be recorded by the teacher and may be used later.

This plugin accepts two input URLs and a secret key (this setting will be set by the administrator of Moodle). The teachers and the students need not have any control over this setting. The secret can be obtained using the following commands: